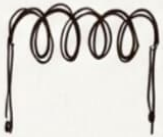
Alternating Current: The Master Notebook



A Complete Study of AC Circuits, Resonance, and Power

Defining Alternating Current

Alternating emf: $\varepsilon = \varepsilon_0 \sin(\omega t)$

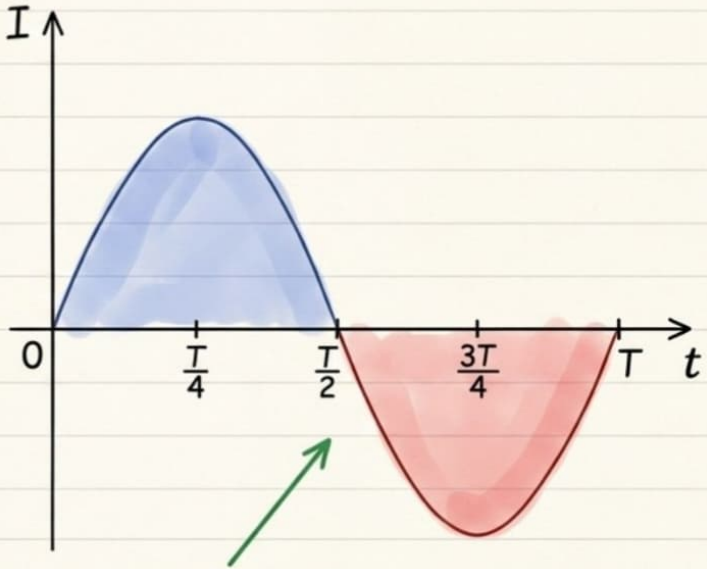
Connected to resistor R:

$$I = \frac{\varepsilon}{R} = \frac{\varepsilon_0 \sin(\omega t)}{R} = I_0 \sin(\omega t)$$

- Real-World Note: Frequency (f) of AC in
Real-World Note: Frequency (f) of AC in
India is 50 Hz, and in the USA is 60 Hz.

AC over a full cycle: Charge flowing in time dt is $dq = I \cdot dt$. Integrating both sides over time 0 to T:

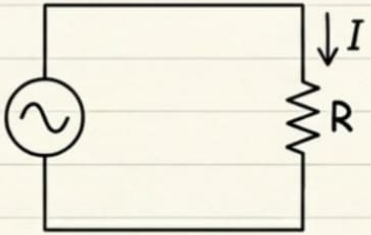
$$q = \int_0^T I_0 \sin(\omega t)\, dt$$

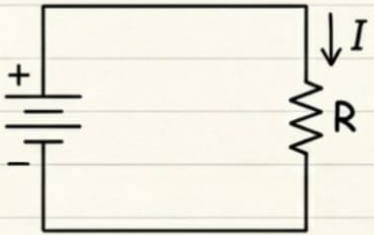


Because the positive and negative areas cancel out, the average current over one full cycle is zero!

The RMS Value of AC (Measuring by Heat)

The value of AC which produces the same heating in a resistor when operated by AC for one full cycle as a steady DC would.





Heat produced by AC in small duration dt: $dH = I^2 R\, dt$

$$dH = I_0^2 \sin^2(\omega t) R\, dt$$

Total Heat (H) for a full cycle (Eq I):

$$H = I_0^2 R \int_0^T \sin^2(\omega t)\, dt$$

Deriving the RMS Multiplier

Step 1 (DC Heat): $H = I_{rms}^2 RT$ (Eq II)

Step 2 (Equating I & II): $I_{rms}^2 RT = I_0^2 R \int_0^T \sin^2(\omega t)\, dt$

Step 3 (Trig Substitution): $I_{rms}^2 = \frac{I_0^2}{T} \int_0^T \left[\frac{1-\cos(2\omega t)}{2}\right] dt$

Use double-angle formula here to make it integrable!

Step 4 (Integration): $I_{rms}^2 = \frac{I_0^2}{2T}[T-0] - \left[\frac{\sin(2\omega t)}{2\omega}\right]_0^T$

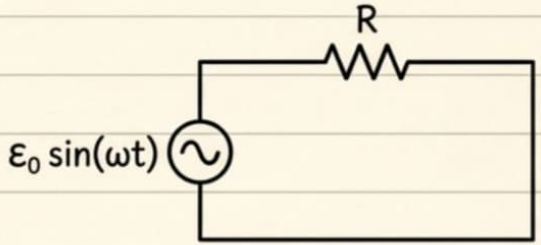
Step 5 (Evaluating Limits): Since $\sin(4\pi) = 0$, the second term vanishes.

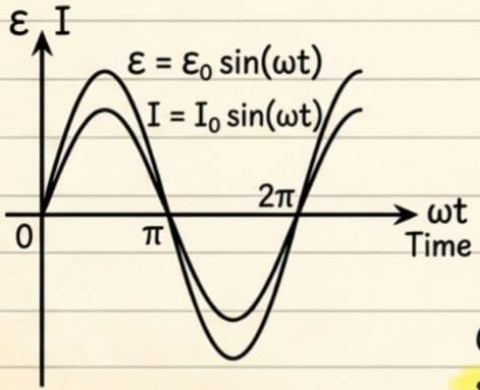
$$I_{rms}^2 = \frac{I_0^2 \cdot T}{2T} = \frac{I_0^2}{2}$$

$$\boxed{I_{rms} = \frac{I_0}{\sqrt{2}}}$$

Similarly, $\varepsilon_{rms} = \frac{\varepsilon_0}{\sqrt{2}}$

The Pure Resistor (Perfect Harmony)





Applied emf: $\varepsilon = \varepsilon_0 \sin(\omega t)$

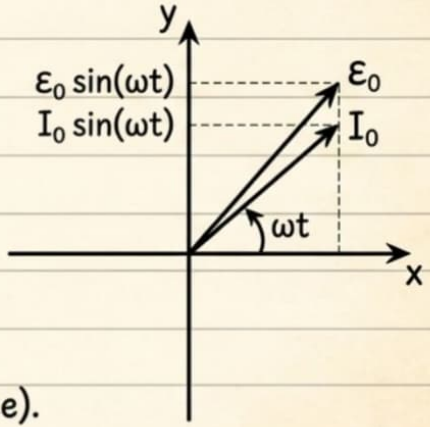
By Kirchhoff's Law:

$\varepsilon = IR \rightarrow \varepsilon_0 \sin(\omega t) = IR$

Resulting Current:

$I = I_0 \sin(\omega t)$

Conclusion: ε and I are in the same phase (no phase difference).



The Pure Inductor (Momentum & Lag)

Applied emf: $\varepsilon = \varepsilon_0 \sin(\omega t)$

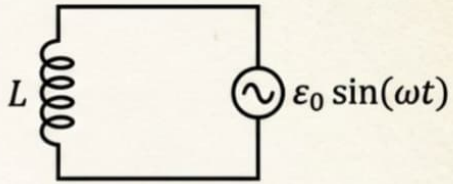
Induced emf: $\varepsilon_L = -L\dfrac{dI}{dt}$

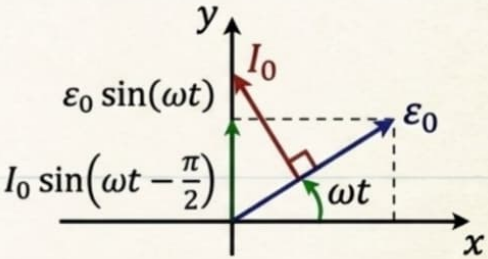
Net emf $= 0 \rightarrow \varepsilon = L\dfrac{dI}{dt} \rightarrow dI = \dfrac{\varepsilon_0}{L}\sin(\omega t)\,dt$

Integrating: $I = \dfrac{\varepsilon_0}{L}\left[\dfrac{-\cos(\omega t)}{\omega}\right] = \dfrac{\varepsilon_0}{\omega L}\sin\left(\omega t - \dfrac{\pi}{2}\right)$

Inductive Reactance (X_L): $X_L = \omega L = 2\pi f L$

Result: ε leads I by a phase difference of $\dfrac{\pi}{2}$.

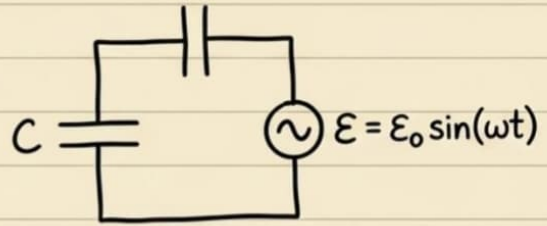




Higher frequencies face more resistance from inductors!

The Pure Capacitor (Pressure & Lead)

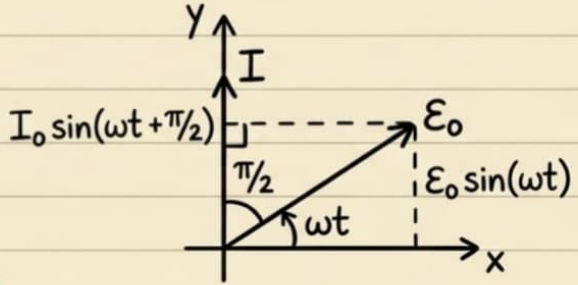
Applied emf: $\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$



Charge: $Q = CV = C \cdot \mathcal{E}_0 \sin(\omega t)$

Current: $I = \dfrac{dQ}{dt} = \dfrac{d[C \cdot \mathcal{E}_0 \sin(\omega t)]}{dt} = C\mathcal{E}_0 \omega \cos(\omega t)$

Rearranging: $I = \dfrac{\mathcal{E}_0}{1/\omega C} \sin(\omega t + \pi/2)$

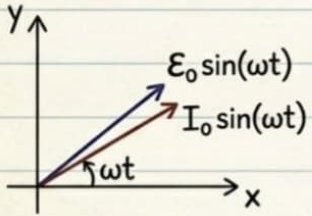
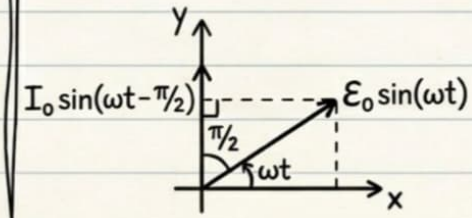
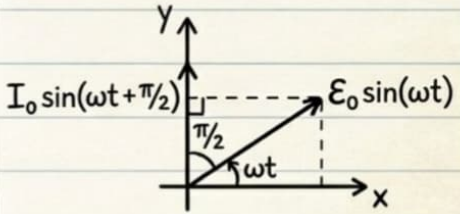


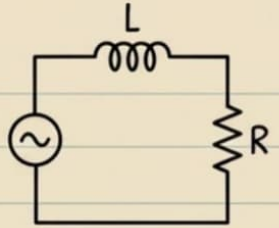
Capacitive Reactance (X_C): $X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi f C}$

Result: I leads $\mathcal{E}$ by a phase difference of $\pi/2$.

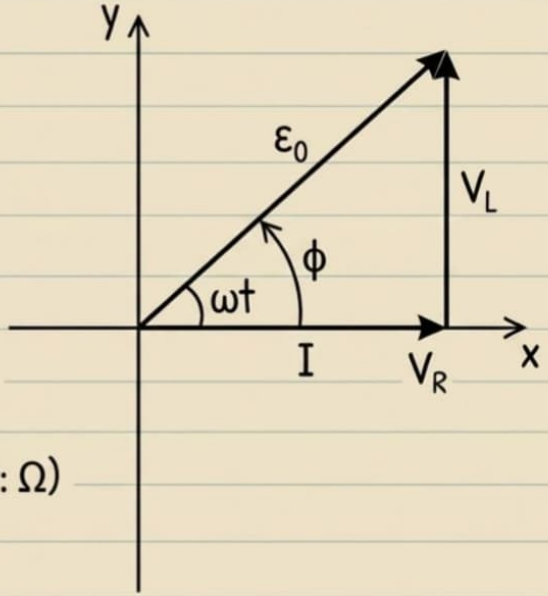
Lower frequencies face more resistance from capacitors!

Synthesis: The Circuit Element DNA

Resistor (R)	Inductor (L)	Capacitor (C)
Resistance: R	Reactance: $X_R = \omega L$	Reactance: $X_C = \frac{1}{\omega C}$
Phase: In Phase ($\phi = 0$)	Phase: I lags ε by $\frac{\pi}{2}$	Phase: I leads ε by $\frac{\pi}{2}$
Proportionality: Independent of f	Proportionality: $X_R \propto f$	Proportionality: $X_C \propto \frac{1}{f}$
		



Combinations: The Series LR Circuit



Potential drops across components:

In Inductor: $V_L = I \cdot X_L$ (V_L leads I by $\pi/2$)

In Resistor: $V_R = I \cdot R$ (Same phase)

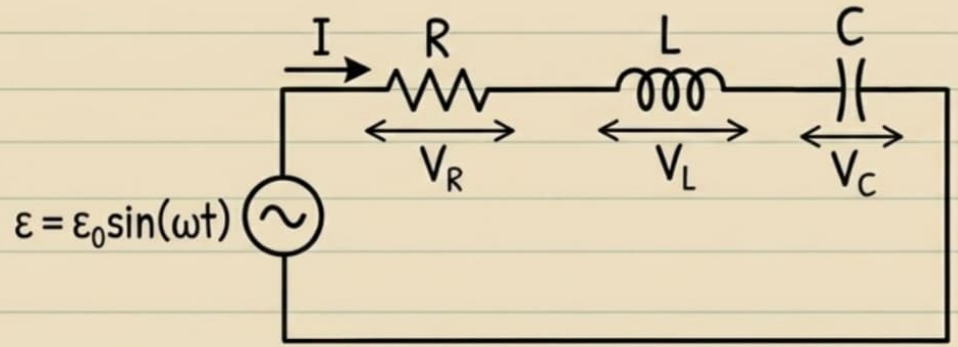
Net emf vector addition: $\varepsilon_0 = \sqrt{V_R^2 + V_L^2} = \sqrt{(I_0 R)^2 + (I_0 X_L)^2}$

Impedance (Z): $\varepsilon_0 = I_0 \sqrt{R^2 + X_L^2} \rightarrow Z = \sqrt{R^2 + X_L^2}$ (SI Unit: Ω)

Phase Angle (ϕ): $\tan\phi = \frac{V_L}{V_R} = \frac{X_L}{R}$

Current equation: $I = I_0 \sin(\omega t - \phi)$

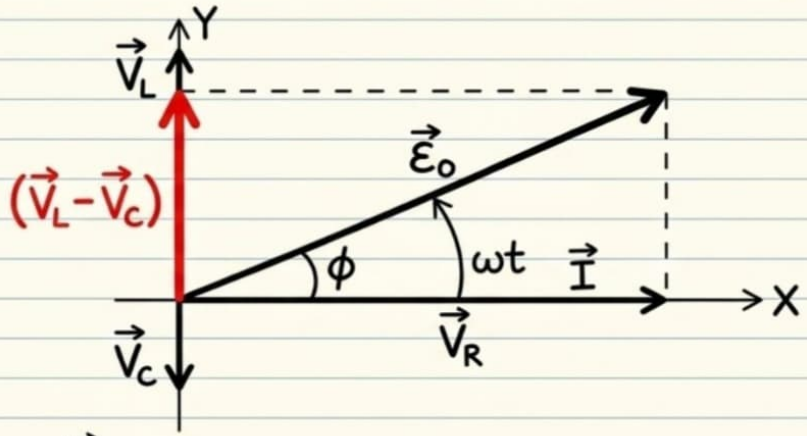
The Ultimate Combination: Series LCR Circuit



The circuit experiences potential drops across all three elements simultaneously:

1. Resistor (R): $V_R = I_0 R$ (Horizontal, in phase)
2. Inductor (L): $V_L = I_0 X_L$ (Pointing up, leading by $\pi/2$)
3. Capacitor (C): $V_C = I_0 X_C$ (Pointing down, lagging by $\pi/2$)

The LCR Phasor: A Vector Battleground



Because $\vec{V}_L$ and $\vec{V}_C$ point in exact opposite directions, they subtract from one another. Assuming $X_L > X_C$, the net vertical vector is $(\vec{V}_L - \vec{V}_C)$.

Net emf: $\mathcal{E}_0 = \sqrt{V_R^2 + (V_L - V_C)^2}$

$$\boxed{\text{Total LCR Impedance: } Z = \sqrt{R^2 + (X_L - X_C)^2}}$$

Current: $I_0 = \dfrac{\mathcal{E}_0}{Z}$

Resonance: Perfect Cancellation

In Resonance, X_L perfectly equals X_C.

Impedance becomes minimal: $Z = \sqrt{R^2 + (X_L - X_L)^2} = R$

(The LCR circuit becomes purely resistive!)

$$\boxed{Z = R}$$

Correction to student notes:
Because $I_0 = \varepsilon_0 / Z$, making Z minimal actually makes I_0 MAXIMAL! This maximum current spike is the definition of resonance.

Deriving Resonant Frequency (f_r):

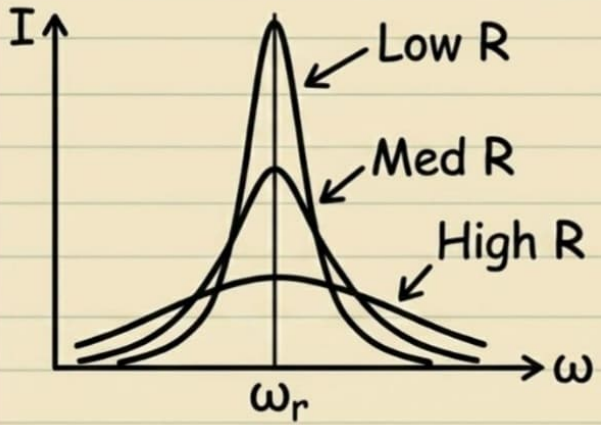
$$X_L = X_C \rightarrow \omega L = \frac{1}{\omega C}$$

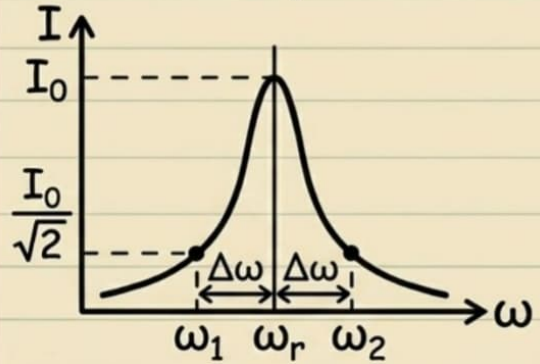
$$\omega^2 = \frac{1}{LC} \rightarrow \omega = \frac{1}{\sqrt{LC}}$$

$$2\pi f_r = \frac{1}{\sqrt{LC}}$$

$$\boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$$

The Q-Factor (Sharpness of Resonance)





Current amplitude peaks at ω_r and falls off in either direction.

The peak is higher for smaller R and flattened for larger R.

Formulas:

$$Q = \frac{\omega_r}{\text{Bandwidth}} = \frac{\omega_r}{2\Delta\omega}$$

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R}$$

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Putting It Into Practice: LCR Example (Part 1)

Given values implied: $\omega = 314$, $L = \frac{1}{10}$ H, $C = 25\ \mu$F, $R = 25\ \Omega$

Step 1: Calculate Reactances

$$X_L = \omega L = 314 \times \frac{1}{10} = 31.4\ \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{10^6}{314 \times 25} = 127.3\ \Omega$$

Net Reactance $= X_C - X_L = 127.3 - 31.4 = 95.9\ \Omega$

Notice here that $X_C > X_L$, meaning the circuit is net-capacitive and current will lead!

Step 2: Calculate Impedance (Z)

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$Z = \sqrt{25^2 + (95.9)^2} = \sqrt{625 + 9196.8} = 99.18\ \Omega$$

Real-World Dynamics: Power in AC Circuits

Definition: Rate of electrical energy consumed in an AC circuit.

Instantaneous Power (P): $P = \varepsilon \cdot I$

Substituting values: $P = [\varepsilon_0 \sin(\omega t)] \cdot [I_0 \sin(\omega t - \phi)]$

Using the trigonometric identity $\underbrace{2\sin A}_{A=\omega t}\,\underbrace{\sin B}_{B=(\omega t - \phi)} = \cos(A-B) - \cos(A+B)$:

$$P = \frac{\varepsilon_0 I_0}{2}\left[\cos\phi - \cos(2\omega t - \phi)\right]$$

$$P = \frac{\varepsilon_0 I_0}{\sqrt{2}\cdot\sqrt{2}}\left[\cos\phi - \cos(2\omega t - \phi)\right]$$

Average Power & The Power Factor

Because the average of the time-dependent function $\cos(2\omega t - \phi)$ over a full cycle is 0:

Average Power (P_{avg}): $$\boxed{P_{\text{avg}} = \frac{\varepsilon_0 I_0}{\sqrt{2}\cdot\sqrt{2}}\cos\phi = \varepsilon_{\text{rms}} I_{\text{rms}}\cos\phi}$$

The Power Factor: $\cos\phi = \dfrac{R}{Z}$

Synthesis application:

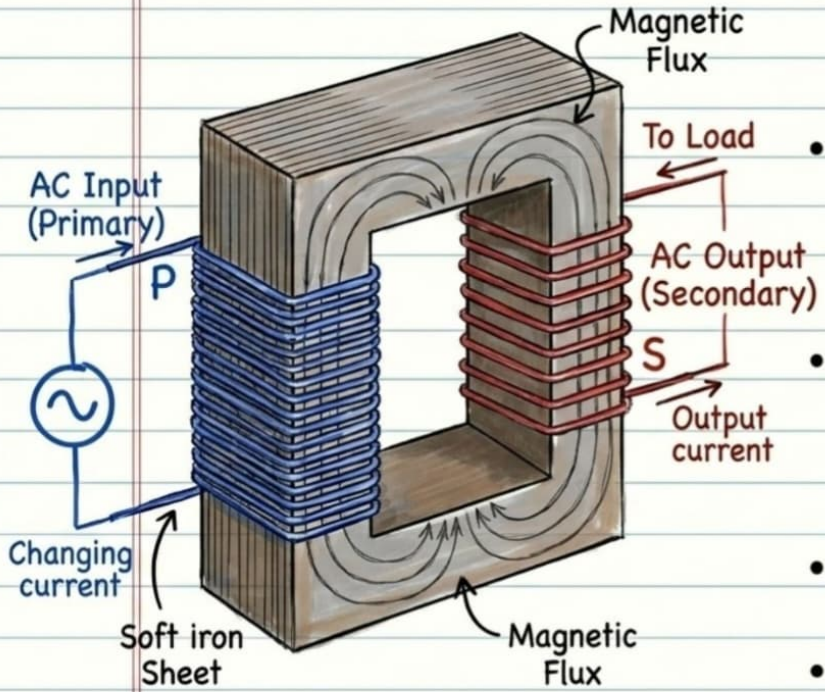
In pure Inductive (L) and pure Capacitive (C) circuits, the phase difference is exactly $\phi = \pi/2$.

Since $\cos(\pi/2) = 0$, the average power consumed is exactly 0!

In L & C, $\phi = \frac{\pi}{2}$, $P = 0$

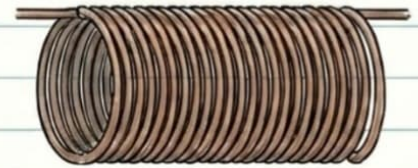
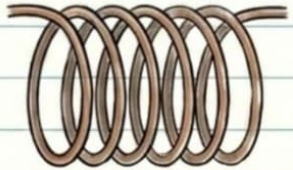
This is why pure inductors and capacitors are considered "wattless" components!

Applying AC at Scale: The Transformer



- **Definition:** An electrical device converting AC at low voltage to AC at high voltage, and vice-versa.
- **Principle:** Works on the principle of Mutual Inductance (changing current passed through one coil sets up an induced emf in the other coil).
- **Construction:** Consists of 2 coils made of insulated Copper wire having a different number of turns, wound on the same soft iron core.
- **Primary Coil:** Input (Electrical energy supplied).
- **Secondary Coil:** Output (Electrical energy drawn)

The Transformer Matrix: Step-Up vs. Step-Down

Step-Up Transformer	Step-Down Transformer
Voltage Change: Increases Input Voltage.	**Voltage Change:** Decreases Input Voltage.
Current Change: Decreases Current.	**Current Change:** Increases Current.
Construction: Secondary coil has more turns than Primary.	**Construction:** Secondary coil has fewer turns than Primary.
	

End of Notes. Master these building blocks, and you've mastered AC circuits.